

01 Scene-Grounding ETC

Sequence in 3D space

Camera Calibration



02 Task-Oriented ETC



Grasp here to pick up the mango

[295, 397]

Affordance



Put the mouse into the box

Planning

Sim

01 Coffee Bag

Real-world setup



Predict the trajectory of:
Pick up the COL beans.

[[<loc_128>, <loc_612>],
[<loc_230>, <loc_590>],
[<loc_333>, <loc_569>],
[<loc_410>, <loc_547>],
[<loc_486>, <loc_533>]]

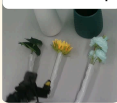
[[128, 612], [230, 590],
[333, 569], [410, 547],
[486, 533]]



Real

02 Flower

Real-world setup



Replace the monstera
with lily.

Generated

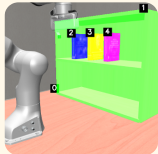


Locate the
lily with
bounding
box.

[551, 11, 17, 931]



03 Action-Aligned Planning ETC



Spatial Understanding

Gripper position to Book 3:

- Front/Back: in front
- Up/Down: slightly above
- Left/Right: centered around it



Object description



Which is the 3-pin tile?

The tile with three dots.



Locate the 3-pin.

[250, 218, 288, 255].



03 BBQ



Real-world setup

Given the
bounding box, is
this region
suitable for a
stable grasp?

Yes.



Yes.



ETC Data